

Python Practice Assignment

Problem 1 – Transaction Processing

Given:

price = 500

quantity = 3

discount = 15%

tax = 18%

Tasks:

1. Calculate total amount before discount.
2. Calculate discount amount.
3. Calculate amount after discount.
4. Add tax to calculate final bill.
5. Print final amount.

Problem 2 – Customer Eligibility Check

Dataset:

age = 28

salary = 45000

loan_default = False

Using **if-else conditions and logical operators**, check:

- Age must be greater than 21
- Salary must be greater than 30000
- Loan default must be **False**

If all conditions satisfy print:

Customer Eligible

Otherwise print:

Customer Not Eligible

Python Practice Assignment

Problem 3 – Data Quality Check

A data pipeline receives a record:

```
email = "user@email.com"
```

Using **if condition** check:

- If "@" exists in email → print "Valid Email"
- Else → print "Invalid Email"

Dataset:

```
record = "cust101|Rahul Sharma|Delhi|rahul@gmail.com"
```

Problem 4 – String Indexing

Print:

1. First character
2. Last character
3. First 6 characters
4. Last 8 characters

Problem 5 – String Slicing

Extract using slicing/split:

- Customer ID
- Customer Name
- City
- Email

Problem 6 – Reverse String

Reverse the following string:

```
customer_id = "CUST10234"
```

Python Practice Assignment

Do it using:

1. Slicing
2. Loop

Problem 7 – String Cleaning

Given:

```
name = " Anil kumar "
```

Tasks:

1. Remove extra spaces
2. Convert to uppercase
3. Replace space with _
4. Check if name starts with "AN"

Expected output:

```
ANIL_KUMAR
```

```
True
```

Dataset:

```
sales = [1200, 900, 1500, 800, 2000]
```

Problem 8 – List Indexing and Slicing

1. Print first element
2. Print last element
3. Print first 3 elements
4. Print last 2 elements

Python Practice Assignment

Problem 9 – List Methods

customers = ["Rahul", "Amit", "John"]

Perform:

1. Add "Priya" using append
2. Insert "Neha" at position 1
3. Remove last element using pop
4. Print updated list

Problem 10 – Data Filtering Using Loop

Dataset:

sales = [1200, 500, 1700, 300, 2100]

Using a **for loop**, create a new list with values **greater than 1000**.

Expected output:

[1200,1700,2100]

Problem 11 – Data Cleaning

Dataset:

names = [" Rahul ", " Amit ", " Priya ", " John "]

Using a **for loop**:

- Remove spaces
- Convert to uppercase
- Store in new list

Expected:

['RAHUL','AMIT','PRIYA','JOHN']

Python Practice Assignment

Problem 12 – Log Processing

Dataset:

```
logs = [  
    "INFO: Job Started",  
    "ERROR: Disk Full",  
    "INFO: Job Running",  
    "ERROR: Memory Issue"  
]
```

Using a **for loop and if condition**, print only **ERROR logs**.

Problem 13 – Stop Pipeline on Critical Error

Dataset:

```
logs = [  
    "INFO: Job Started",  
    "INFO: Loading Data",  
    "CRITICAL: Database Down",  
    "INFO: Job Completed"  
]
```

Using a **for loop**:

- If "CRITICAL" appears, print message and **break the loop**

Expected output:

INFO: Job Started

INFO: Loading Data

Critical Error Found. Stopping Pipeline

Problem 14 – Skip Invalid Records

Dataset:

```
records = ["cust101", "", "cust102", "", "cust103"]
```

Python Practice Assignment

Using **continue**, skip empty records and print valid ones.

Expected output:

```
cust101  
cust102  
cust103
```

Problem 16 – Customer Record

Create a dictionary:

```
customer = {  
    "id": "cust101",  
    "name": "Rahul",  
    "city": "Delhi",  
    "age": 30  
}
```

Print:

- customer name
- city

Problem 17 – Dictionary Methods

Given:

```
employee = {  
    "id": 101,  
    "name": "Anirudh",  
    "department": "Data Engineering"  
}
```

Tasks:

1. Add new key "salary" = 80000
 2. Update department to "Analytics"
 3. Print all keys
 4. Print all values
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Python Practice Assignment

Problem 18 – Loop Through Dictionary

Dataset:

```
customer = {  
    "id": "cust101",  
    "name": "Rahul",  
    "city": "Delhi",  
    "email": "rahul@gmail.com"  
}
```

Using a **for loop**, print:

```
id : cust101  
name : Rahul  
city : Delhi  
email : rahul@gmail.com
```